

INTRODUCTION

The dataset used in this project is from the UCI Machine Learning Repository, which records annual spending by 440 business clients across six categories, which are Fresh, Milk, Grocery, Frozen, Detergents & Paper, and Delicassen, alongside Channel (Horeca vs. Retail) and Region (Lisbon, Oporto, Other). We explore five key questions: channel spending differences, channel prediction, regional separation, how Frozen spending is related to other product categories, and whether customer groupings align with channel or region.

Descriptive analysis shows Retail customers spend far more on Grocery, Milk, and Detergents and Paper, whereas Horeca clients spend more in Fresh and Frozen. A stepwise logistic model retains only Grocery and Detergents_Paper as predictors of Retail. To test regional differentiation, we performed cross-validated quadratic discriminant analysis and found overall accuracy is around 38 percent. Stepwise selection retained Fresh, Milk, Detergents_Paper, and Delicassen as the significant predictors in the final model for Frozen spending. Clustering produces three segments that perfectly map to Channel (one Retail, one Horeca, one mixed) but not to Region, underscoring that segmentation and strategy should be channel-focused rather than region-based.

Task 1

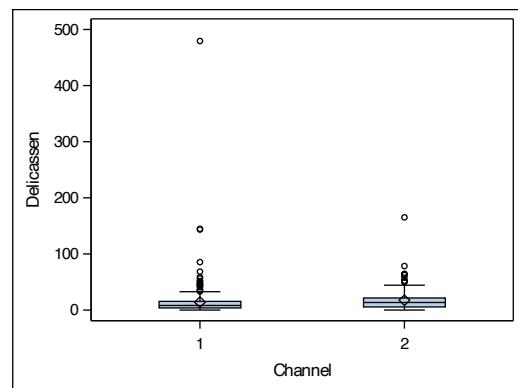
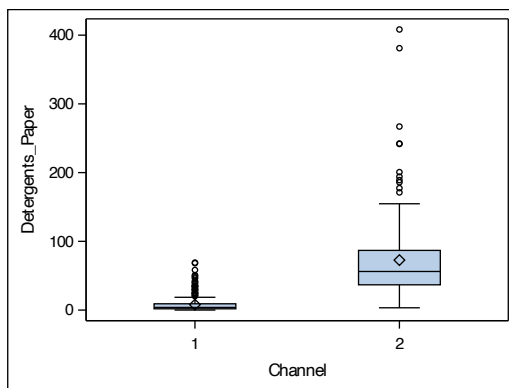
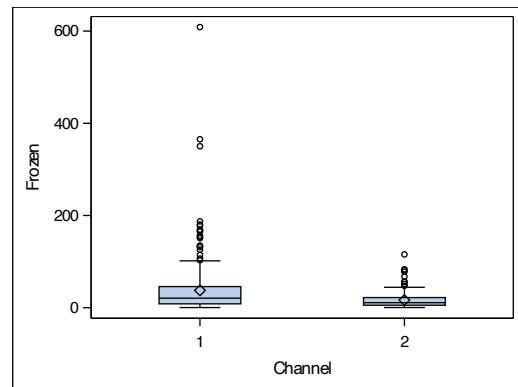
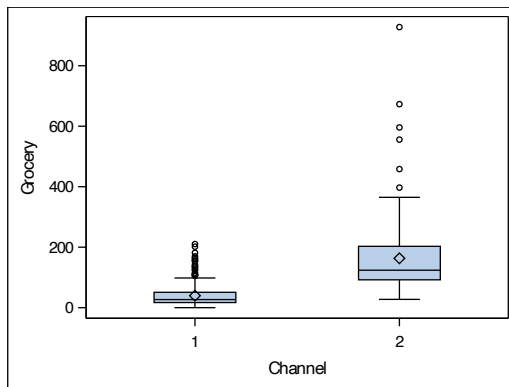
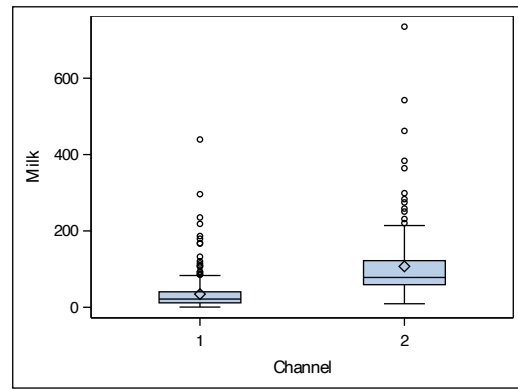
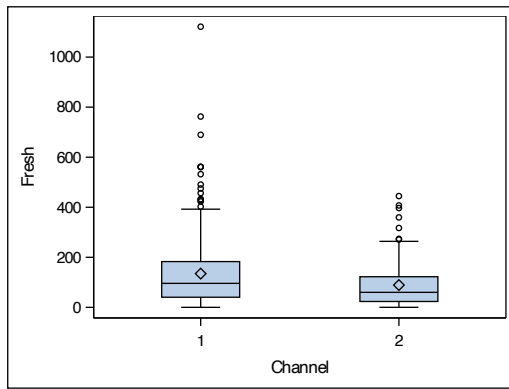
Retail customers (channel 2) spend substantially more on grocery (mean=163.23), milk (107.17), and detergents_paper (72.70), while Horeca customers (channel 1) spend more on fresh (134.76) and frozen (37.48) items. Fresh spending in the Horeca channel exhibits the greatest variability (SD=138.32), whereas in the retail channel, grocery spending shows the largest dispersion (SD=122.67). In most categories, the channel with the higher average spend also shows the most extreme outliers. For delicassen spending, the two channels have similar average levels, but Horeca shows the single largest outliers, indicating that some hotels, restaurants and cafes make occasional very large specialty purchases. Shapiro-Wilk tests confirm strong right-skewness in every category (all p-values less than 0.0001), providing evidence that none of the spending variables are normally distributed (see A, Task 1 Appendix). Overall, these results underscore that retail outlets focus on packaged goods (high spending on grocery, milk, detergents), while HORECA businesses incur larger, more variable costs for fresh and frozen goods, with occasional large specialty orders in delicassen.

Channel=1

Variable	N	Mean	Std Dev	Minimum	Maximum
Fresh	298	134.7556040	138.3168750	0.0300000	1121.51
Milk	298	34.5172483	43.5216557	0.5500000	439.5000000
Grocery	298	39.6213758	35.4551339	0.0300000	210.4200000
Frozen	298	37.4825168	56.4391250	0.2500000	608.6900000
Detergents_Paper	298	7.9056040	11.0409367	0.0300000	69.0700000
Delicassen	298	14.1595638	31.4742692	0.0300000	479.4300000

Channel=2

Variable	N	Mean	Std Dev	Minimum	Maximum
Fresh	142	89.0432394	89.8771475	0.1800000	444.6600000
Milk	142	107.1650000	96.7963135	9.2800000	734.9800000
Grocery	142	163.2285211	122.6731809	27.4300000	927.8000000
Frozen	142	16.5261268	18.1280366	0.3300000	115.5900000
Detergents_Paper	142	72.6950704	62.9108970	3.3200000	408.2700000
Delicassen	142	17.5343662	19.5379705	0.0300000	165.2300000

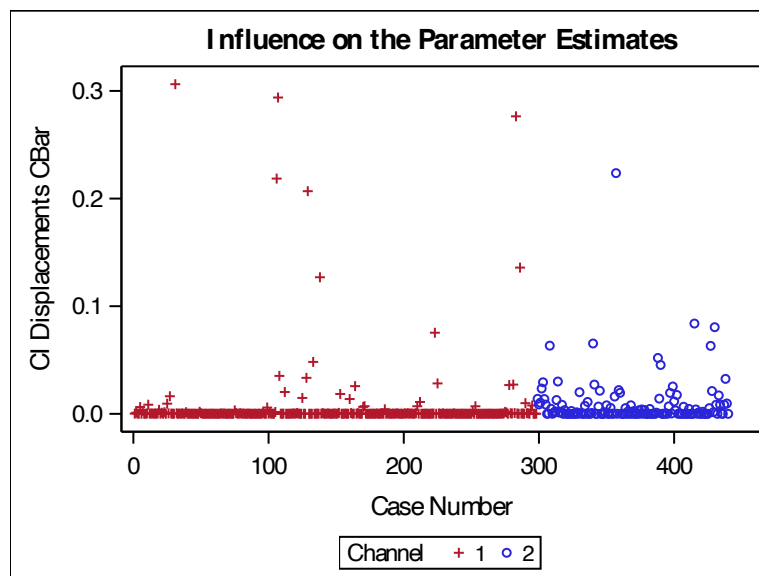


Task 2

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-3.9880	0.3749	113.1395	<.0001
Grocery	1	0.0129	0.00454	8.0901	0.0045
Detergents_Paper	1	0.0928	0.0123	57.1691	<.0001

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Grocery	1.013	1.004	1.022
Detergents_Paper	1.097	1.071	1.124

Hosmer and Lemeshow Goodness-of-Fit Test		
Chi-Square	DF	Pr > ChiSq
19.5435	8	0.0122

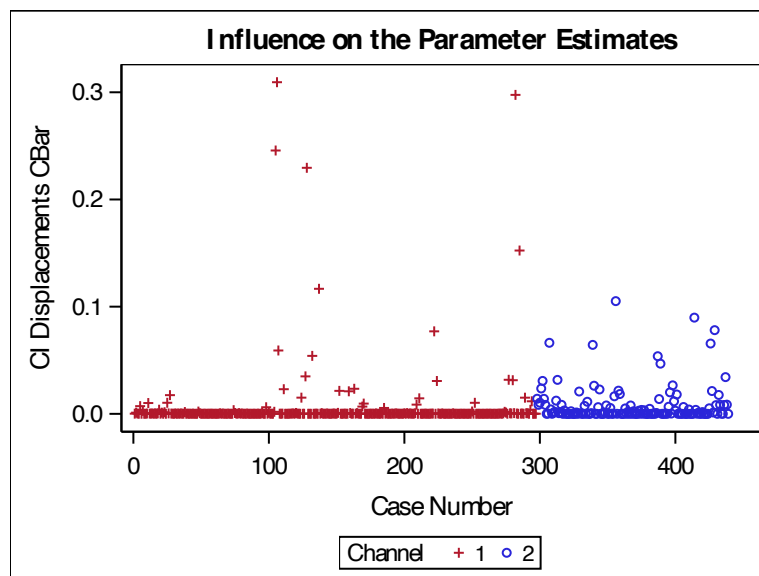


First, we fitted a logistic regression. With stepwise selection, we find that Grocery and Detergents_Paper are the only significant predictors for whether a business is a Retail business. However, the Hosmer–Lemeshow p-value is below 0.05, meaning that there is a lack of fit. We should consider removing a high Cbar value and refit the model.

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-4.1180	0.3905	111.1902	<.0001
Grocery	1	0.0152	0.00465	10.6821	0.0011
Detergents_Paper	1	0.0925	0.0124	55.7935	<.0001

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Grocery	1.015	1.006	1.025
Detergents_Paper	1.097	1.071	1.124

Hosmer and Lemeshow Goodness-of-Fit Test		
Chi-Square	DF	Pr > ChiSq
16.2769	8	0.0386

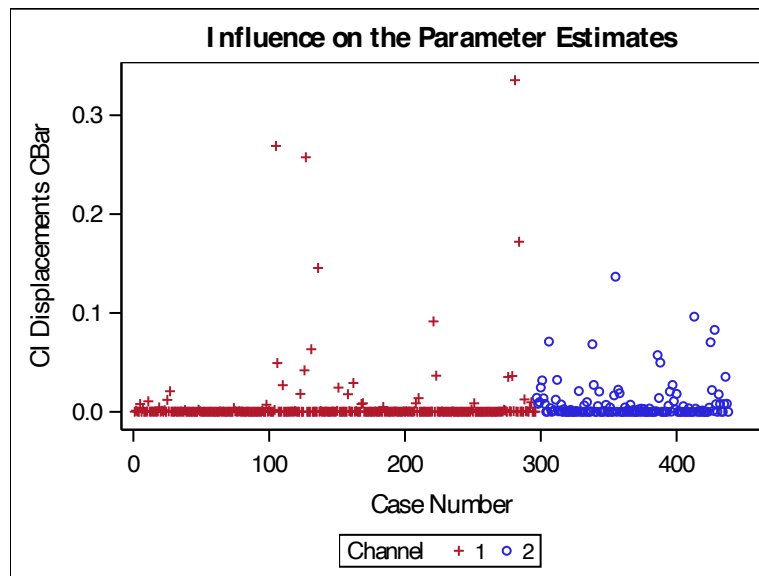


The Hosmer–Lemeshow p-value improved, but it is still below 0.05. We continue to remove another high Cbar value and refit the model again.

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-4.2054	0.4033	108.7266	<.0001
Grocery	1	0.0148	0.00472	9.8570	0.0017
Detergents_Paper	1	0.0991	0.0129	58.6670	<.0001

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Grocery	1.015	1.006	1.024
Detergents_Paper	1.104	1.077	1.133

Hosmer and Lemeshow Goodness-of-Fit Test		
Chi-Square	DF	Pr > ChiSq
13.4739	8	0.0966



Now, the results are very good. The Hosmer–Lemeshow p-value is greater than 0.05, indicating there is no lack of fit. The best model includes only Grocery and Detergents_Paper. Both Grocery and Detergents_Paper parameters are positive, suggesting that those who spend more in Grocery and Detergents_Paper have higher log odds of being a Retail channel. The odds ratios confirm this. Each additional unit of Grocery spending increases the odds of being Retail by 1.5 percent. Since the confidence interval (1.006, 1.025) does not include 1, this effect is statistically significant. Also, for every one unit increase in Detergents Paper, the odds of being a Retail channel are multiplied by around 1.104. The confidence interval (1.077, 1.133) excludes 1, suggesting a significant effect. Although there are some CBar values higher than the majority of values, none are extreme enough to raise concerns about undue influence.

Task 3

The test of Homogeneity of Within Covariance Matrices was statistically significant, with p-value less than 0.0001, meaning that the assumption of equal covariances was violated. Not all p-values from the multivariate test of group mean differences was significant, except from Roy's test. This suggests that there is no strong evidence that the mean spending profiles differ across regions. The cross-validation results were not great. Customers from Lisbon were correctly classified 33.8 percent of the time, Oporto a mere 8.5 percent, and Other regions 56 percent. The overall cross-validation error rate was high, approximately 53 percent. This indicates that spending alone provide minimal discriminatory power across regions and strongly supports the distributor's hypothesis that regions cannot be distinguished by purchase behavior.

The Method for Selecting Variables is STEPWISE			
Total Sample Size	440	Variable(s) in the Analysis	6
Class Levels	3	Variable(s) Will Be Included	0
		Significance Level to Enter	0.05
		Significance Level to Stay	0.05

Number of Observations Read	440
Number of Observations Used	440

Class Level Information				
Region	Variable Name	Frequency	Weight	Proportion
1	1	77	77.0000	0.175000
2	2	47	47.0000	0.106818
3	3	316	316.0000	0.718182

Stepwise Selection: Step 1

Statistics for Entry, DF = 2, 437				
Variable	R-Square	F Value	Pr > F	Tolerance
Fresh	0.0052	1.13	0.3235	1.0000
Milk	0.0017	0.38	0.6855	1.0000
Grocery	0.0025	0.55	0.5772	1.0000
Frozen	0.0048	1.06	0.3466	1.0000
Detergents_Paper	0.0036	0.79	0.4550	1.0000
Delicassen	0.0033	0.72	0.4898	1.0000

No variables can be entered.

No further steps are possible.

Test of Homogeneity of Within Covariance Matrices

Chi-Square	DF	Pr > ChiSq
374.911944	42	<.0001

Since the Chi-Square value is significant at the 0.1 level, the within covariance matrices will be used in the discriminant function.

Reference: Morrison, D.F. (1976) Multivariate Statistical Methods p252.

Multivariate Statistics and F Approximations					
S=2 M=1.5 N=215					
Statistic	Value	F Value	Num DF	Den DF	Pr > F
Wilks' Lambda	0.96580439	1.26	12	864	0.2351
Pillai's Trace	0.03426593	1.26	12	866	0.2386
Hotelling-Lawley Trace	0.03533354	1.27	12	668.92	0.2317
Roy's Greatest Root	0.03313606	2.39	6	433	0.0277
NOTE: F Statistic for Roy's Greatest Root is an upper bound.					
NOTE: F Statistic for Wilks' Lambda is exact.					

Classification Summary for Calibration Data: WORK.WHOLESALE
Cross-validation Summary using Quadratic Discriminant Function

Number of Observations and Percent Classified into Region				
From Region	1	2	3	Total
1	26 33.77	4 5.19	47 61.04	77 100.00
2	14 29.79	4 8.51	29 61.70	47 100.00
3	128 40.51	11 3.48	177 56.01	316 100.00
Total	168 38.18	19 4.32	253 57.50	440 100.00
Priors	0.175	0.10682	0.71818	

Error Count Estimates for Region				
	1	2	3	Total
Rate	0.6623	0.9149	0.4399	0.5295
Priors	0.1750	0.1068	0.7182	

Task 4

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Deviance	433	214.7144	0.4959
Scaled Deviance	433	472.6807	1.0916
Pearson Chi-Square	433	107.7800	0.2489
Scaled Pearson X2	433	237.2712	0.5480
Log Likelihood		-1669.9552	
Full Log Likelihood		-1669.9552	
AIC (smaller is better)		3355.9104	
AICC (smaller is better)		3356.2445	
BIC (smaller is better)		3388.6046	

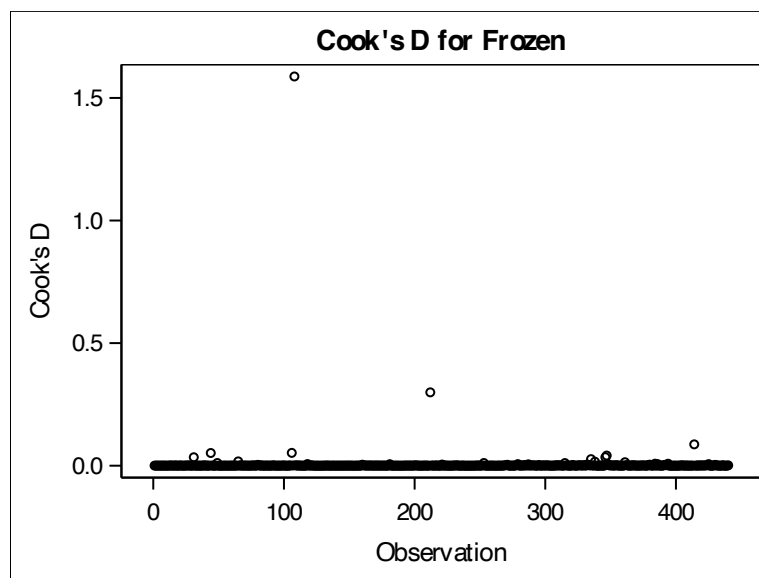
Analysis Of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept	1	2.1413	0.0630	2.0178	2.2647	1155.44	<.0001
Fresh	1	0.0006	0.0003	-0.0000	0.0012	3.61	0.0574
Milk	1	0.0006	0.0008	-0.0009	0.0022	0.66	0.4163
Grocery	1	-0.0019	0.0010	-0.0039	0.0001	3.40	0.0652
Frozen	1	0.0255	0.0013	0.0229	0.0281	372.08	<.0001
Detergents_Paper	1	0.0020	0.0020	-0.0019	0.0059	1.03	0.3108
Delicassen	1	0.0000	0.0019	-0.0037	0.0037	0.00	0.9826
Scale	1	2.2014	0.1387	1.9458	2.4907		

Note: The scale parameter was estimated by maximum likelihood.

LR Statistics For Type 1 Analysis				
Source	2*LogLikelihood	DF	Chi-Square	Pr > ChiSq
Intercept	-3880.4912			
Fresh	-3805.2109	1	75.28	<.0001
Milk	-3804.9690	1	0.24	0.6228
Grocery	-3779.3382	1	25.63	<.0001
Frozen	-3340.9935	1	438.34	<.0001

LR Statistics For Type 1 Analysis				
Source	2*LogLikelihood	DF	Chi-Square	Pr > ChiSq
Detergents_Paper	-3339.9109	1	1.08	0.2981
Delicassen	-3339.9104	1	0.00	0.9826

LR Statistics For Type 3 Analysis			
Source	DF	Chi-Square	Pr > ChiSq
Fresh	1	3.72	0.0539
Milk	1	0.68	0.4102
Grocery	1	3.32	0.0683
Frozen	1	423.42	<.0001
Detergents_Paper	1	1.02	0.3124
Delicassen	1	0.00	0.9826



We modeled frozen food sales using Gamma GLM with a log link to accommodate its right-skewed distribution. The scaled deviance of 1.1684 is not too far from 0, so we do not have to worry about overdispersion. In the parameter estimates, not all predictors are significant. Cook's distance plot suggests the presence of an unduly influential point violating the rule of thumb. We should remove this point and refit the model.

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Deviance	433	505.9378	1.1684
Scaled Deviance	433	506.7059	1.1702
Pearson Chi-Square	433	609.3817	1.4073
Scaled Pearson X2	433	610.3068	1.4095
Log Likelihood		-1872.4070	

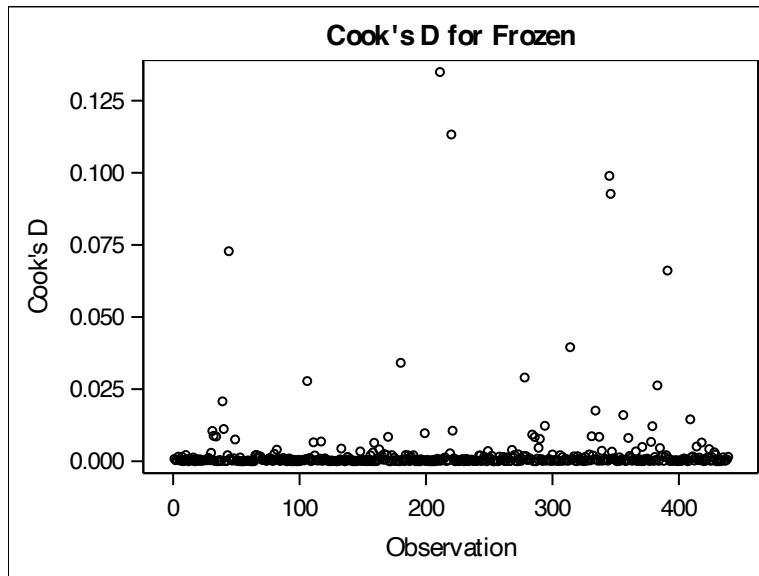
Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Full Log Likelihood		-1872.4070	
AIC (smaller is better)		3758.8140	
AICC (smaller is better)		3759.0738	
BIC (smaller is better)		3787.4055	

Analysis Of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept	1	2.9222	0.0811	2.7633	3.0812	1298.16	<.0001
Fresh	1	0.0023	0.0005	0.0014	0.0032	26.27	<.0001
Milk	1	0.0026	0.0013	0.0002	0.0051	4.42	0.0354
Grocery	1	-0.0010	0.0016	-0.0041	0.0021	0.37	0.5428
Detergents_Paper	1	-0.0062	0.0028	-0.0117	-0.0006	4.76	0.0292
Delicassen	1	0.0119	0.0039	0.0042	0.0195	9.26	0.0023
Scale	1	1.0015	0.0595	0.8914	1.1253		

Note: The scale parameter was estimated by maximum likelihood.

LR Statistics For Type 1 Analysis				
Source	2*LogLikelihood	DF	Chi-Square	Pr > ChiSq
Intercept	-3852.4015			
Fresh	-3785.7792	1	66.62	<.0001
Milk	-3785.4580	1	0.32	0.5709
Grocery	-3764.7097	1	20.75	<.0001
Detergents_Paper	-3755.6766	1	9.03	0.0027
Delicassen	-3744.8140	1	10.86	0.0010

LR Statistics For Type 3 Analysis			
Source	DF	Chi-Square	Pr > ChiSq
Fresh	1	30.76	<.0001
Milk	1	4.65	0.0311
Grocery	1	0.37	0.5439
Detergents_Paper	1	4.73	0.0296
Delicassen	1	10.86	0.0010



Grocery is the only insignificant predictor in the parameter estimates. We should exclude it from the final model.

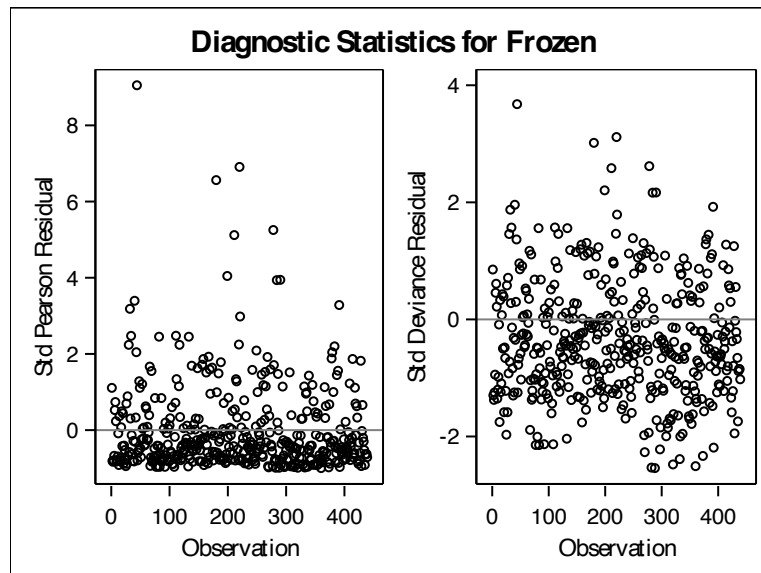
Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Deviance	434	506.3057	1.1666
Scaled Deviance	434	506.7439	1.1676
Pearson Chi-Square	434	604.3431	1.3925
Scaled Pearson X2	434	604.8662	1.3937
Log Likelihood		-1872.5911	
Full Log Likelihood		-1872.5911	
AIC (smaller is better)		3757.1823	
AICC (smaller is better)		3757.3767	
BIC (smaller is better)		3781.6893	

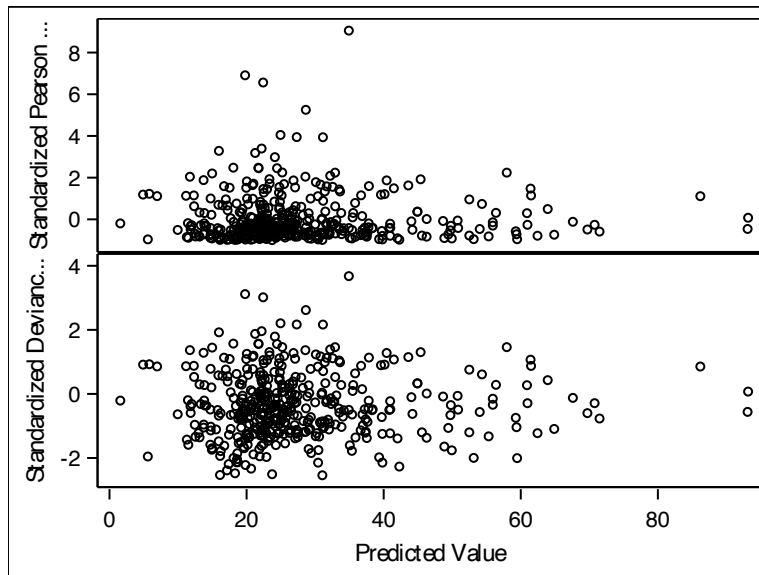
Analysis Of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept	1	2.9143	0.0802	2.7571	3.0715	1319.90	<.0001
Fresh	1	0.0023	0.0005	0.0014	0.0032	25.79	<.0001
Milk	1	0.0024	0.0011	0.0001	0.0046	4.19	0.0407
Detergents_Paper	1	-0.0076	0.0014	-0.0105	-0.0048	28.45	<.0001
Delicassen	1	0.0116	0.0039	0.0040	0.0192	8.92	0.0028
Scale	1	1.0009	0.0595	0.8908	1.1245		

Note: The scale parameter was estimated by maximum likelihood.

LR Statistics For Type 1 Analysis				
Source	2*LogLikelihood	DF	Chi-Square	Pr > ChiSq
Intercept	-3852.4015			
Fresh	-3785.7792	1	66.62	<.0001
Milk	-3785.4580	1	0.32	0.5709
Detergents_Paper	-3755.6776	1	29.78	<.0001
Delicassen	-3745.1823	1	10.50	0.0012

LR Statistics For Type 3 Analysis			
Source	DF	Chi-Square	Pr > ChiSq
Fresh	1	30.54	<.0001
Milk	1	4.31	0.0378
Detergents_Paper	1	24.11	<.0001
Delicassen	1	10.50	0.0012





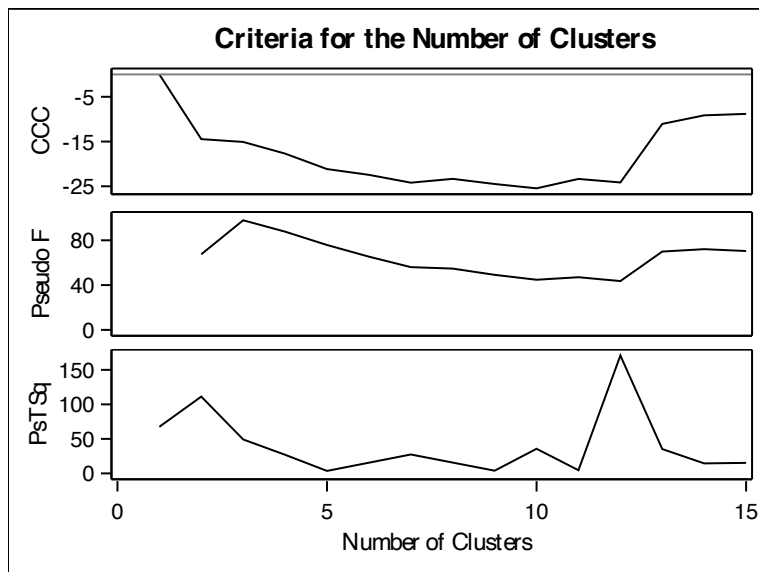
Our final model includes Fresh, Milk, Detergents Paper, and Delicassen as predictors of Frozen food sales. Parameter estimates show that all predictors are highly significant with p-values less than 0.05. Type 3 analysis confirms this, while type 1 disagrees, showing that Milk is not significant. But AIC has decreased, which is a good sign. Cook's distance plot also shows no concerning influential point. The plots for standardized Pearson and deviance residuals versus predicted values seem fine and do not appear to have any trend, meaning that the final model fits the data well. The coefficients for Fresh, Milk, Detergents Paper, and Delicassen are $\exp(0.0023)=1.0023$, $\exp(0.0024)=1.0024$, $\exp(-0.0076)=0.9924$, and $\exp(0.0116)=1.0117$ respectively. The model tells us that:

- For each additional 100 units spent on Fresh, expected Frozen spending increases by 0.23 percent.
- For each additional 100 units spent on Milk, expected Frozen spending increases by 0.24 percent.
- Each extra 100 units on Detergents and Paper lower the expected spending on Frozen by 0.76 percent.
- And each extra 100 units on Delicassen increase the the expected spending on Frozen by 1.17 percent.

Task 5

Based on the correlation-based principal component analysis, we keep 2 components. For component 1, all coefficients are positive, but largest values are in Milk, Grocery, and Detergents Paper, which are what Retail customers usually spend more on. For component 2, there is a contrast of spending from Retail customers and Horeca customers (see A, Task 5 Appendix).

In terms of clusters, we see that the CCC values are negative at the first 15 components, and the pseudo F statistic is relatively high at 3 clusters. Also, the pseudo t-squared shows a large jump from 3 to 2 cluster. So, 3 is our best choice for number of clusters.



Here we can do a simple comparison. We see that component 1 is highest in cluster 2 and pretty high in cluster 3. In general, those two clusters spend heavily on Milk, Grocery, and Detergents_Paper. Cluster 1 has a slight negative average component 1. In contrast, component 2 is highest in cluster 3 and negative in other clusters.

We say that cluster 1 exhibits moderate spending across all goods. Cluster 2 is a packaged-goods-heavy segment, with exceptionally high Grocery, Milk, and Detergents_Paper spend, but low Fresh and Frozen purchases. Finally, cluster 3 includes a single extreme outlier, who is a large Horeca customer. They spend much more money than everyone else on Fresh, Frozen, and Delicassen.

CLUSTER=1

Variable	N	Mean	Std Dev	Minimum	Maximum
Prin1	434	-0.1251791	1.2201608	-1.4350761	5.8039727
Prin2	434	-0.0191542	1.1079911	-2.4521404	8.9952480

CLUSTER=2

Variable	N	Mean	Std Dev	Minimum	Maximum
Prin1	5	9.3077276	1.9910389	7.5399982	12.6386155
Prin2	5	-1.1240575	1.9527471	-3.3760460	1.5059229

CLUSTER=3

Variable	N	Mean	Std Dev	Minimum	Maximum
Prin1	1	7.7890740	.	7.7890740	7.7890740
Prin2	1	13.9332239	.	13.9332239	13.9332239

Looking at the table of cluster by Channel, cluster 2 seem to be 100 percent Retail, cluster 3 100 percent Horeca, while cluster 1 a mix of both (68 percent Horeca and 32 Retail). On the other hand, the table of cluster by Region does not show much relation. Each cluster was dominated by Other region, with only minimal representation from Lisbon or Oporto in any group.

These results confirm that spending-based segmentation aligns strongly with channel rather than region. Retail customers cluster together through their heavy purchases of packaged goods (Milk, Grocery, and Detergents_Paper), while Horeca clients from their own group driven by Fresh and Frozen goods. Also, the majority of businesses fall into a moderate cluster (cluster 1).

Table of CLUSTER by Channel			
CLUSTER	Channel		
Frequency Percent	1	2	Total
1	297 67.50	137 31.14	434 98.64
2	0 0.00	5 1.14	5 1.14
3	1 0.23	0 0.00	1 0.23
Total	298 67.73	142 32.27	440 100.00

Table of CLUSTER by Region				
CLUSTER	Region			
Frequency Percent	1	2	3	Total
1	77 17.50	46 10.45	311 70.68	434 98.64
2	0 0.00	1 0.23	4 0.91	5 1.14
3	0 0.00	0 0.00	1 0.23	1 0.23
Total	77 17.50	47 10.68	316 71.82	440 100.00

CONCLUSION

In summary, Retail customers focus on packaged goods, which are Grocery and Detergents_Paper, the only significant predictors in our logistic model, while Horeca clients lead in Fresh and Frozen foods. Regional classification fails (overall cross-validation error rate was high, around 53 percent) so geography offers no spend-based segmentation. Frozen sales are driven by Fresh, Milk, Detergents_Paper, and Delicassen, revealing cross-category demand patterns. Finally, business spending clearly segments by channel, not by region. The distributor might want to tailor marketing strategies by channel rather than region-based approaches.

APPENDIX

Task 1

A

Variable:
Fresh
Channel = 1

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.78416	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.165019	Pr > D	<0.0100
Cramer-von Mises	W-Sq	2.458404	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	14.34703	Pr > A-Sq	<0.0050

Variable:
Fresh
Channel = 2

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.840451	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.1614	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.009225	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	6.069043	Pr > A-Sq	<0.0050

Variable:
Milk
Channel = 1

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.584983	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.218075	Pr > D	<0.0100
Cramer-von Mises	W-Sq	5.236625	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	28.97205	Pr > A-Sq	<0.0050

Variable:
Milk
Channel = 2

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.667952	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.212016	Pr > D	<0.0100
Cramer-von Mises	W-Sq	2.227375	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	12.19403	Pr > A-Sq	<0.0050

Variable:
Grocery

Channel = 1

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.782827	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.177945	Pr > D	<0.0100
Cramer-von Mises	W-Sq	3.251138	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	18.47584	Pr > A-Sq	<0.0050

Variable:
Grocery
Channel = 2

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.731408	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.166254	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.460783	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	8.561163	Pr > A-Sq	<0.0050

Variable:
Frozen
Channel = 1

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.558747	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.254725	Pr > D	<0.0100
Cramer-von Mises	W-Sq	5.650255	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	30.3912	Pr > A-Sq	<0.0050

Variable:
Frozen
Channel = 2

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.744048	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.19123	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.680992	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	9.573526	Pr > A-Sq	<0.0050

Variable:
Detergents_Paper
Channel = 1

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.645859	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.237827	Pr > D	<0.0100
Cramer-von Mises	W-Sq	6.261785	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	33.59951	Pr > A-Sq	<0.0050

Variable:
Detergents_Paper
Channel = 2

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.755783	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.188371	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.566687	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	8.729854	Pr > A-Sq	<0.0050

Variable:
Delicassen
Channel = 1

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.288902	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.326743	Pr > D	<0.0100
Cramer-von Mises	W-Sq	9.894898	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	51.40086	Pr > A-Sq	<0.0050

Variable:
Delicassen
Channel = 2

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.688381	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.185149	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.544331	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	8.932423	Pr > A-Sq	<0.0050

Task 5

A

Eigenvalues of the Correlation Matrix				
	Eigenvalue	Difference	Proportion	Cumulative
1	2.64497357	0.94238960	0.4408	0.4408
2	1.70258397	0.96251920	0.2838	0.7246
3	0.74006477	0.17633455	0.1233	0.8479
4	0.56373023	0.27805388	0.0940	0.9419
5	0.28567634	0.22270523	0.0476	0.9895
6	0.06297111		0.0105	1.0000

Eigenvectors						
	Prin1	Prin2	Prin3	Prin4	Prin5	Prin6
Fresh	0.042884	0.527932	0.812257	-.236686	0.048683	0.036025
Milk	0.545118	0.083168	-.060388	-.087190	-.826579	0.038040
Grocery	0.579256	-.146088	0.108384	0.105987	0.314999	-.721745
Frozen	0.051189	0.611278	-.178386	0.768683	0.027932	0.015637
Detergents_Paper	0.548640	-.255233	0.136192	0.171744	0.339640	0.685894
Delicassen	0.248682	0.504207	-.523904	-.552065	0.314701	0.075134

